

Monte Carlo and Resampling Methods Applied to Weak Gravitational Lensing Mass Reconstruction

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Abstract

We present a Monte Carlo and resampling study of uncertainty quantification for a weak gravitational lensing mass reconstruction pipeline based on the Kaiser–Squires (KS) algorithm. The forward operator mapping convergence to shear is implemented in R and validated analytically against the SMPy production implementation. Simple Monte Carlo, antithetic variable sampling, and importance sampling are applied to characterize reconstruction error under shape noise. Parametric bootstrap confidence intervals, both percentile and bias-corrected-and-accelerated (BCa), are constructed for two scalar summaries of the reconstructed mass map. A coverage simulation confirms that 95% BCa intervals achieve empirical coverage of 97.5%. A power analysis determines that approximately 24,400 source galaxies are required to detect 1% multiplicative shear bias at 5-sigma significance with 500 Monte Carlo replicates per condition. The antithetic design is found to be degenerate for the L^2 reconstruction error, a consequence of the estimator’s symmetry under sign reversal of the noise, illustrating a fundamental condition on variance reduction methods not always emphasized in textbook treatments.

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1. Introduction

Weak gravitational lensing is among the most powerful observational probes of cosmological large-scale structure. Background galaxy images are coherently distorted by intervening mass along the line of sight, producing a measurable shear field from which the projected mass density, the convergence κ , can be reconstructed. The standard linear inversion, known as Kaiser–Squires reconstruction (Kaiser & Squires, 1993), maps the observed shear back to convergence via a Fourier-domain inversion of the lensing kernel.

From a statistical standpoint the problem is a regularized linear inverse problem: a known linear forward operator F maps the unknown κ to the observable shear γ , and the reconstruction is obtained from noisy shear measurements subject to the irreducible intrinsic ellipticity scatter of source galaxies (shape noise). Several core statistical questions arise naturally: how should uncertainty in the reconstructed mass map be quantified; what confidence intervals are appropriate for scalar summaries; and how many source galaxies are required to detect systematic errors in the shear measurement pipeline?

This project applies the Monte Carlo and resampling methods of MA 551 to answer these questions within a fully controlled simulation environment. Section 2 describes the lensing forward model, the reconstruction algorithm, and the noise model. Section 3 presents the simulation study covering simple Monte Carlo, antithetic variables, importance sampling, and bootstrap confidence intervals. Section 4 develops the power analysis. Section 5 concludes with connections to the broader weak lensing literature.

2. Methodology

2.1. The Kaiser–Squires Forward–Inverse System

The convergence $\kappa(\boldsymbol{\theta})$ and shear $\gamma = \gamma_1 + i\gamma_2$ are related through the lensing potential ψ , which satisfies $\nabla^2\psi = 2\kappa$. In Fourier space, the shear is obtained from κ via the Kaiser–Squires kernel $D(\mathbf{k})$ (Kaiser & Squires, 1993):

$$\hat{\gamma}(\mathbf{k}) = D(\mathbf{k})\hat{\kappa}(\mathbf{k}), \quad D(\mathbf{k}) = \frac{k_1^2 - k_2^2 + 2ik_1k_2}{k_1^2 + k_2^2},$$

with $D(\mathbf{0}) = 0$ by convention (the DC mode is unobservable). The inverse, which recovers κ from γ , is:

$$\hat{\kappa}(\mathbf{k}) = \overline{D(\mathbf{k})}\hat{\gamma}(\mathbf{k})$$

since $|D(\mathbf{k})| = 1$ for $\mathbf{k} \neq \mathbf{0}$. Optional Tikhonov regularization replaces the denominator with $|D|^2 + \lambda$, shrinking high-frequency modes.

²<https://github.com/GeorgeVassilakis/SMPy>

The implementation in R was validated analytically against the SMPy² KaiserSquiresMapper: both compute $(k_1^2 - k_2^2)\hat{G}_1 + 2k_1k_2\hat{G}_2)/k^2$ for the E-mode, which is algebraically identical to $\text{Re}[\overline{D}\hat{\gamma}]$ (Notes #11, #12). A 10-test analytical suite confirmed noiseless round-trip error below 10^{-10} , $|D(\mathbf{k})| = 1$ to machine precision, and B-mode purity below 10^{-10} for a physical convergence field.

2.2. Noise Model and Simulation Setup

We simulate source galaxy shape noise at the pixel level. With N_{gal} galaxies uniformly distributed over an $N \times N$ grid, the per-pixel shape noise standard deviation is (Bartelmann & Schneider, 2001):

$$\sigma_{\text{pix}} = \frac{\sigma_e}{\sqrt{N_{\text{gal}}/N^2}},$$

where $\sigma_e = 0.26$ per component is the intrinsic ellipticity dispersion. The observed shear at each pixel is:

$$\gamma_{\text{obs}[i,j]} = (1 + m)\gamma_{\text{true}[i,j]} + c + \varepsilon[i,j], \quad \varepsilon[i,j] \sim \mathcal{N}(0, \sigma_{\text{pix}}^2),$$

where m is a multiplicative bias and c an additive bias injected for the power analysis. This is the standard Gaussian shape noise model used in analytical weak lensing studies (Kaiser & Squires, 1993).

We use a 32×32 pixel grid with pixel scale 0.1 arcmin, giving a 3.2 arcmin field. The true convergence is a Gaussian cluster profile $\kappa_{\text{true}}(\theta) = 0.3 \exp(-|\theta|^2 / 2\sigma_\ell^2)$ with $\sigma_\ell = 0.5$ arcmin. With $N_{\text{gal}} = 5000$ galaxies, $\sigma_{\text{pix}} = 0.118$ and the shear signal reaches $|\gamma| \approx 0.10$, giving a per-pixel SNR of approximately 0.85.

Reconstruction statistic. Two scalar summaries of the KS reconstruction are used throughout: the peak convergence $T_{\text{peak}} = \max(\hat{\kappa})$ and the aperture mass $T_{\text{ap}} = \sum_{|\theta| \leq r_{\text{ap}}} \hat{\kappa}(\theta), \delta\theta^2$ with $r_{\text{ap}} = 0.8$ arcmin. The aperture mass is preferred for inference because it averages over a spatial region rather than taking the maximum of a noisy field; this makes it far more stable under the shape noise model.

DC ambiguity. The KS inversion always zeros the DC Fourier mode, so $E[\hat{\kappa}] = 0$ but κ_{true} has a non-zero mean. The aperture mass of the noiseless KS reconstruction ($T_{\text{ap}}^{\text{KS}} = 0.2504$) differs from the true aperture mass ($T_{\text{ap}}^{\text{true}} = 0.3461$) by 9.6%, reflecting the unrecoverable mass-sheet degeneracy (Bartelmann & Schneider, 2001).

3. Simulation Study

3.1. Simple Monte Carlo

We estimate $E[L_2(m = 0)]$, the mean relative L^2 reconstruction error at zero bias, using $B = 500$ independent noise realizations (Notes #12). The L^2 error is computed on the interior (4-

Summary	True value	MC mean	MC SD
L_2 error	0 (m=0)	1.521	0.045
Peak $\hat{\kappa}$	0.297	0.488	0.051
Aperture mass	0.250*	0.251	0.015

* Noiseless KS aperture mass; true-kappa value is 0.346 (DC offset 9.6%).

Table 1: Simple Monte Carlo results ($N_{\text{gal}} = 5000$, $B = 500$).

pixel border removed) after subtracting the mean from both reconstructed and true fields to remove the DC offset. Results are summarized in Table 1.

The aperture mass estimate is nearly unbiased relative to the noiseless KS target. The peak kappa estimate shows substantial positive bias (0.488 vs. 0.297 true peak): the maximum of a 1024-pixel noisy field is a noise spike, not the cluster center.

3.2. Antithetic Variable Sampling

Antithetic sampling pairs each noise draw ε with its negation $-\varepsilon$, exploiting negative correlation to reduce Monte Carlo variance (Notes #13). The variance reduction is:

$$\text{Var}(\hat{I}_{\text{anti}}) = \frac{\sigma^2}{2N} (1 + \text{Cor}(g(\varepsilon), g(-\varepsilon))).$$

For a monotone integrand the correlation is negative, so variance is reduced. Results for the three summaries at $B = 500$ pairs are:

- **Aperture mass:** +100% variance reduction (correlation = -1 , exact)
- **Peak kappa:** -4% variance reduction (correlation slightly positive)
- **L_2 error:** -100% variance reduction (degenerate case)

The degenerate result for L_2 error is analytically exact, not a numerical artifact. The reconstruction is linear: $\hat{\kappa}(\varepsilon) = F^{\{-1\}}(F\kappa_{\text{true}} + \varepsilon)$, so the mean-corrected L_2 error satisfies $L_2(\varepsilon) = \|\varepsilon_{\text{KS}}\| / \|\kappa_{\text{true}}\|$ where $\varepsilon_{\text{KS}} = F^{\{-1\}}\varepsilon$ is the noise propagated through the KS inverse. Since this depends only on $\|\varepsilon\|$, we have $L_2(\varepsilon) = L_2(-\varepsilon)$ exactly, so antithetic pairs yield identical values. Averaging them doubles the sample without reducing variance, halving the effective sample size.

Aperture mass achieves perfect negative correlation because it is a spatially integrated linear functional of $\hat{\kappa}$, and $\sum \hat{\kappa}(\varepsilon) = -\sum \hat{\kappa}(-\varepsilon)$ (mean zero, linear in ε).

This illustrates the condition stated in Notes #13: antithetic sampling requires that g be **monotone**. The L_2 error, being even in ε , violates this condition. This finding would directly affect survey design decisions that rely on variance reduction efficiency.

3.3. Importance Sampling

We estimate the bias-integrated reconstruction error $E_{p[L_2(m)]} = \int L_2(m)p(m)dm$, where $p(m) = \mathcal{N}(0, 0.05^2)$ is a Gaussian prior on multiplicative bias (Notes #12, #13). The proposal is $q(m) = \text{Uniform}(-0.15, 0.15)$, giving self-normalized importance weights $w_i = \frac{p(m_i)}{q(m_i)}$.

Using $n_{\text{IS}} = 50$ proposal draws (each evaluated with $B' = 100$ MC replicates), the IS estimate is $\hat{I}_{\text{IS}} = 1.518$ (SE = 0.193), compared to the uniform grid estimate of 1.520. The small difference reflects that the error function is nearly flat over the range of m values supported by the prior, so IS offers little advantage here. This is itself an informative finding: the reconstruction error is insensitive to small biases at this galaxy density, meaning the bias signal is buried in shape noise.

3.4. Bootstrap Confidence Intervals

We construct parametric bootstrap confidence intervals for the aperture mass using $B = 500$ bootstrap replicates at $N_{\text{gal}} = 5000$ (Notes #6, #7). In the parametric bootstrap, fresh shape noise is drawn from the known $\mathcal{N}(0, \sigma_{\text{pix}}^2)$ distribution for each replicate, consistent with the known noise model.

The BCa interval adjusts for bias and skewness via the bias-correction factor $\hat{z}_0 = \Phi^{\{-1\}}\left(\frac{1}{B} \sum_b \mathbb{1}(\hat{T}_b < \hat{T})\right)$ and the acceleration $\hat{a}$, estimated via column-jackknife on the shear grid (deleting one column of $N = 32$ pixels at a time, requiring N reconstructions rather than N^2):

$$\hat{a} = \frac{\sum_j \left(|(T)_{\text{jack}} - \hat{T}_{(j)} \right)^3}{6 \left\{ \sum_j \left(|(T)_{\text{jack}} - \hat{T}_{(j)} \right)^2 \right\}^{\frac{3}{2}}}.$$

Results are shown in Table 2.

The peak kappa BCa is degenerate: every bootstrap replicate draws fresh noise, so the bootstrap peak (driven by noise spikes) always exceeds the noiseless reference value. This makes $\hat{z}_0 = \Phi^{\{-1\}}(0) = -\infty$ and the interval undefined. This is not a coding error; it correctly identifies that the peak of a noise-dominated field cannot be bootstrapped with a parametric resampling scheme that adds noise to a noiseless reference. The aperture mass is well-

Statistic	Obs.	Bias	SE	95% BCa CI
Peak $\hat{\kappa}$	0.251	+0.237	0.051	degenerate
Aperture mass	0.250	+0.000	0.015	[0.220, 0.278]

Table 2: Bootstrap results ($B = 500$, $N_{\text{gal}} = 5000$). Peak kappa BCa is degenerate ($\hat{z}_0 = -\infty$) because all bootstrap replicates exceed the noiseless observed value.

behaved: bias is negligible (+0.0003), and the BCa correction is small ($\hat{z}_0 = -0.045$, $\hat{a} = 0.006$), indicating near-symmetry of the bootstrap distribution.

The acceleration $\hat{a} \approx 0.006$ near zero reflects the near-linear relationship between shear and aperture mass. For non-linear statistics or when the estimator's standard error changes rapidly with the parameter, $\hat{a}$ is substantially nonzero and the BCa correction becomes important.

3.5. Bootstrap Coverage Study

To assess whether the bootstrap intervals achieve their nominal coverage, we run $n_{\text{outer}} = 200$ independent outer Monte Carlo replicates (Notes #6). For each outer replicate, one noisy shear observation is generated, a 95% BCa and percentile CI are computed, and coverage against the noiseless KS aperture mass is checked.

Results: both BCa and percentile intervals achieve 97.5% empirical coverage at nominal 95%, with mean CI width 0.059. The slight over-coverage (97.5% vs. 95%) is consistent with the conservative behavior of the parametric bootstrap when σ_{pix} is known exactly and the shape noise distribution is exactly Gaussian. The near-identical width of BCa and percentile CIs (0.0585 vs. 0.0586) confirms that the acceleration correction is small, as expected for this near-linear estimator.

4. Power Analysis

We ask: how many source galaxies N_{gal} are required to detect multiplicative bias $|m| = 0.01$ at 5-sigma significance, using a two-sample Welch t -test on B independent aperture mass replicates per condition?

Why unpaired? A paired design, sharing noise draws between $m = 0$ and $m = 0.01$, is degenerate for aperture mass, which is a linear statistic. The paired difference $D_b = T_{\text{ap}(m=0.01)} - T_{\text{ap}(m=0)} = 0.01 \times T_{\text{ap}}^{\text{KS}(m=0)}$ is exactly constant across replicates, giving $\text{sd}(D) = 0$ and $\text{NCP} = \infty$. The unpaired design preserves the natural noise contribution to the variance of each condition.

Analytic power. The effect size is $\delta = |m| \times T_{\text{ap}}^{\text{KS}} = 0.01 \times 0.2504 = 0.002504$. The aperture mass standard deviation scales as $\sigma_{\text{ap}} \propto \frac{1}{\sqrt{N_{\text{gal}}}}$ (shape noise model):

$$\sigma_{\text{ap}(N_{\text{gal}})} = \sigma_{\text{ap,ref}} \sqrt{\frac{N_{\text{gal,ref}}}{N_{\text{gal}}}},$$

with $\sigma_{\text{ap,ref}} = 0.0146$ at $N_{\text{ref}} = 5000$. The two-sample non-centrality parameter is:

$$\text{NCP} = \frac{\delta}{\sqrt{2} \frac{\sigma_{\text{ap}(N_{\text{gal}})}}{\sqrt{B}}}.$$

Power results for $B = 500$ replicates per condition are shown in Table 3.

N_{gal}	NCP	Power	σ_{ap}
1,000	1.21	< 0.001	0.033
2,000	1.71	0.001	0.023
5,000	2.70	0.008	0.015
10,000	3.83	0.096	0.010
20,000	5.41	0.610	0.007
50,000	8.55	> 0.999	0.005

5-sigma threshold: $\alpha = 2.87 \times 10^{\{-7\}}$. $B = 500$ replicates per condition.

Table 3: Power curve for detecting $|m| = 0.01$ at 5-sigma significance.

The power curve shows a rapid transition from negligible power at $N_{\text{gal}} = 5000$ to near-certainty at $N_{\text{gal}} = 50000$. The minimum galaxy count for 80% power is $N_{\text{gal}} \approx 24\{, \}400$ (obtained via uniroot). An MC verification at $N_{\text{gal}} = 5000$, $B = 500$ gives NCP = 2.30 and power = 0.002, in agreement with the analytic prediction (NCP = 2.70, power = 0.008; the 15% gap in NCP reflects finite-sample fluctuation at $B = 500$).

This result has direct survey design implications: detecting sub-percent shear calibration errors at cosmological precision requires tens of thousands of galaxies per field, a finding consistent with requirements from Stage-IV weak lensing surveys (Bartelmann & Schneider, 2001).

5. Conclusions

We have applied the full toolkit of MA 551 (simple Monte Carlo, antithetic variables, importance sampling, parametric bootstrap, BCa confidence intervals, and power analysis) to a weak gravitational lensing mass reconstruction problem. Several findings merit emphasis.

The antithetic design is degenerate for the L_2 reconstruction error, a consequence of the estimator being an even function of the noise. This illustrates the theoretical requirement (Notes #13) that antithetic sampling requires monotonicity of the integrand. The same linearity that makes aperture mass tractable for bootstrap inference renders it degenerate for paired power testing.

The BCa confidence interval for aperture mass achieves 97.5% empirical coverage at nominal 95%, and is nearly identical to the percentile interval because the acceleration parameter $\hat{a} \approx 0.006$ is close to zero. This reflects the near-linearity of the aperture mass functional. The BCa interval would differ substantially from the percentile interval for non-linear statistics such as peak kappa, but peak kappa is itself problematic for bootstrap inference when noise dominates the signal.

The DC mode ambiguity of the KS reconstruction (specifically the mass-sheet degeneracy) accounts for a 9.6% deficit between the reconstructed and true aperture mass. This irreducible offset must be accounted for when comparing bootstrap intervals to the physical truth; the correct target for bootstrap inference is the noiseless reconstruction, not the true convergence map.

The power analysis shows that detecting $|m| = 0.01$ at 5-sigma with 500 replicates requires approximately 24,400 source galaxies. This is achievable with modern wide-field surveys but highlights that per-field shear calibration at the sub-percent level remains statistically demanding.

Future work will incorporate GalSim-simulated galaxy images with realistic PSF convolution and metacalibration shear response correction, placing this statistical analysis in a physically complete pipeline.

Bibliography

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6. Appendix: R Code

Selected key functions are reproduced below. Full source is in the accompanying project repository.

Kaiser-Squires inverse (lensing.R):

```
ks_inverse <- function(gamma1, gamma2, grid, lambda = 0) {
  N <- grid$N_pix
  g1_ft <- fft(gamma1)
  g2_ft <- fft(gamma2)
  gamma_ft <- g1_ft + 1i * g2_ft # preserves N x N dims
  KX <- outer(rep(1, N), grid$kx)
  KY <- outer(grid$ky, rep(1, N))
  K2 <- KX^2 + KY^2; K2[1, 1] <- 1
  D_re <- (KX^2 - KY^2) / K2
  D_im <- (2 * KX * KY) / K2
  D_complex <- D_re + 1i * D_im; D_complex[1, 1] <- 0 + 0i
  Dmod2 <- D_re^2 + D_im^2; Dmod2[1, 1] <- 1
  kappa_ft <- Conj(D_complex) / (Dmod2 + lambda) * gamma_ft
  kappa_ft[1, 1] <- 0 + 0i
  Re(fft(kappa_ft, inverse = TRUE)) / N^2
}
```

BCa confidence interval (bootstrap.R):

```
bca_ci <- function(boot_out, gamma_true, grid, stat_fn = peak_kappa,
  conf = 0.95, lambda = 0, N_gal = NULL) {
  t_boot <- boot_out$t_boot; t_obs <- boot_out$t_obs
  alpha <- 1 - conf
  ci_perc <- quantile(t_boot, c(alpha/2, 1 - alpha/2), names = FALSE)
  z0 <- qnorm(mean(t_boot < t_obs))
  # Column jackknife for acceleration
  N <- grid$N_pix
  g1 <- gamma_true$gamma1; g2 <- gamma_true$gamma2
  t_jack <- numeric(N)
  for (j in seq_len(N)) {
    g1j <- g1; g1j[, j] <- 0
    g2j <- g2; g2j[, j] <- 0
    t_jack[j] <- stat_fn(ks_inverse(g1j, g2j, grid, lambda))
  }
  tj <- mean(t_jack)
  a <- sum((tj - t_jack)^3) / (6 * sum((tj - t_jack)^2)^1.5)
  p_lo <- pnorm(z0 + (z0 + qnorm(alpha/2)) / (1 - a*(z0 + qnorm(alpha/2))))
  p_hi <- pnorm(z0 + (z0 + qnorm(1-alpha/2)) / (1 - a*(z0 + qnorm(1-alpha/2))))
  list(percentile = ci_perc,
    bca = quantile(t_boot, c(p_lo, p_hi), names = FALSE),
    z0 = z0, a = a)
}
```

Analytic power for two-sample test (run_all.R):

```

unpaired_power_analytic <- function(N_gal, B, delta, sd_ref, N_gal_ref,
                                     alpha = 2.87e-7) {
  sd_ng <- sd_ref * sqrt(N_gal_ref / N_gal)
  se <- sqrt(2) * sd_ng / sqrt(B)
  ncp <- delta / se
  t_crit <- qnorm(1 - alpha / 2)
  pnorm(ncp - t_crit) + pnorm(-ncp - t_crit)
}
# Minimum N_gal via root-finding:
ng_80 <- uniroot(function(ng) unpaired_power_analytic(ng, B=500, ...) - 0.80,
                  c(1e3, 1e7))$root

```